Optimizing Instructions and Demonstrations for Multi-Stage Language Model Programs

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HarnessRSI 64 ★★

evidence: retained

Paper

1 What changes, what is kept

Search artifact: each module's prompt
instructions + demonstration sets, jointly
Fixed: model weights and program control flow

1 Bootstrap demonstrations
Run training inputs; keep module traces when the final task metric passes a threshold.

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2 Propose grounded instructions
LM reads training-data / program summaries, successful demos and prompting tips.

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3 Choose a joint configuration
Multivariate TPE selects instructions and demo sets across the program's modules.

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4 Score; update the surrogate
Run on a sampled training minibatch. End-to-end metric guides the next selection.

▼

5 Retain the best full evaluation
Periodically score promising combinations §4.3: train; App. F: validation (unresolved).

Signal: labeled task metrics, not an LLM judge. GPT-4 supplies Conditional answer-type annotations with author supervision.

Models: Llama-3-8B task LM; GPT-3.5 / GPT-4 proposer; GPT-4o teacher for ScoNe / HoVer.

2 Highest test mean on 5 of 7

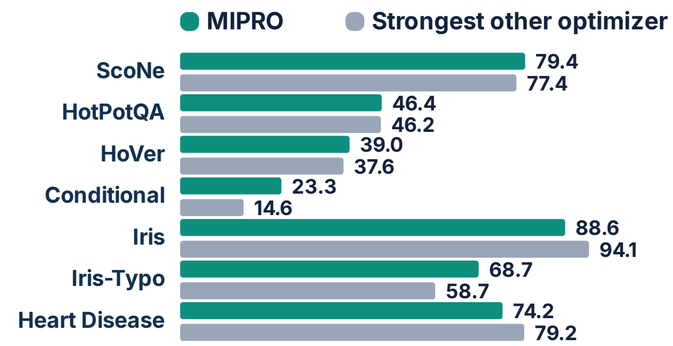


Table 2 · five-run means · Llama-3-8B. HoVer: 3-supporting-fact subset (mixed hops). Task metrics differ; no pooled score, SDs / CIs or seed IDs.

Comparator: Bayesian Bootstrap on ScoNe / HotPotQA / HoVer; 0-Shot MIPRO on Conditional; Bootstrap RS on Iris / Iris-Typo / Heart Disease.

4 Where the gain is absent or thin

- **Gain concentrated:** largest positive gaps are Conditional 23.3 vs 14.6 and Iris-Typo 68.7 vs 58.7.
- **HotPotQA:** 46.4 vs 46.2; Bootstrap RS 45.8 is also statistically comparable. Five top means ≠ five significant wins.
- **Loses on Iris** (88.6 vs 94.1) and **Heart Disease** (74.2 vs 79.2).
- Conditional starts with handwritten task rules. No frozen-prompt cross-task or cross-model transfer study.

3 Controls, access and budgets

Which prompt components get optimized?

Optimizer	Deployed components
Bootstrap RS	Demos · random search
Bayesian Bootstrap	Demos · TPE search
0-Shot MIPRO	Instructions only
Module-Level OPRO	Instructions · score history
MIPRO	**Instructions + demos jointly**

OPRO −G still sees examples + score history in its meta-prompt; it lacks extra grounding.

Grounding is not uniformly helpful

OPRO with / without grounding: ScoNe 73.5 / 76.1; HoVer 32.5 / 25.7. Not a full-MIPRO ablation.

Learning to propose: instruction-only

++ vs instruction-only MIPRO: ScoNe 75.7 / 71.5; HotPotQA 39.3 / 36.8; HoVer 32.6 / 33.1. Temperature unresolved: proposer (§4.4) vs Llama (B.4).

Partial budget match, not all-in cost

Full-eval trials: 50 HotPotQA / ScoNe (~300 minibatch trials); 30 Iris / Heart / Conditional; 20 HoVer. Baselines get these budgets. Total tokens / LM calls, dollars and separate tuning budgets: unreported.

Selection data vs final evaluation

Main: train search, dev iteration, test final. App. F: dev/validation selection; split unresolved ≠ test leakage. Iris / Heart: no dev/iteration. Defaults: intuition; split unspecified. No overlap audit.

Takeaway

Search demonstrations as well as instructions. Joint optimization can help conditional rules and repair a misspecified prompt, but demonstration-only search wins on Iris and Heart Disease. Evaluation-budget matching does not establish equal total optimization cost.

Our read · HarnessRSI

Score 64 · ★★ · evidence: retained

Reports: held-out test, five-run means, significance, partial budget match.

Not reported: SDs / CIs, total optimizer cost, leakage audit, cross-task transfer.